

Surface tension correlations from symbolic regression

$$T_r = \frac{T}{T_c}, \quad F_1 = 1 - T_r$$

Equation 1

Complexity: 1

Loss: 27.3254

$$\sigma = 7.889$$

Equation 2

Complexity: 3

Loss: 0.769627

$$\sigma = (1 - T_r)57.56$$

Equation 3

Complexity: 5

Loss: 0.00190957

$$\sigma = ((1 - T_r)32.42)^{1.304}$$

Equation 4

Complexity: 7

Loss: 0.000264346

$$\sigma = (((1 - T_r)35.15) - 0.1518)^{1.265}$$

Equation 5

Complexity: 9

Loss: 7.44465e-5

$$\sigma = (((1 - T_r)36.5) - 0.3268)^{1.25} + 0.1521$$

Equation 6

Complexity: 11

Loss: 2.78138e-5

$$\sigma = ((Pr - 1.056)Pr) + ((1 - T_r)35.7)^{1.249}$$

Equation 7

Complexity: 13

Loss: 1.94167e-5

$$\sigma = ((Pr - 1.018)Pr) + (((1 - T_r)36.12) - 0.03666)^{1.245}$$

Equation 8

Complexity: 15

Loss: 9.11303e-6

$$\sigma = ((PrPr) + 2.786)((1 - T_r)30.26) + ((3.234)^{Pr} - 3.246))$$

Equation 9

Complexity: 17
Loss: 2.45689e-6

$$\sigma = ((Pr(Pr + 0.2141)) + 3.776)((2.735)^{Pr} - 2.748) + ((1 - T_r)22.54))$$

Equation 10

Complexity: 19
Loss: 2.43787e-6

$$\sigma = (((Pr(Pr + 0.2398)) + 3.779)((2.72)^{Pr} + (((1 - T_r)22.55) - 2.661))) - 0.3633$$